Chapter 2
Health Economics

Preview

- The origins of health economics and the work health economists do are described.
- The parts of health economics that are useful to infection control professionals are highlighted.
- A particular interpretation of health economics that makes decision-making easier is described.

2.1 Origins and Content of Health Economics

Health economics has grown out of mainstream economics. Its birthday was in 1963, when Kenneth Arrow published a paper on how perfectly competitive markets and public sector agencies (i.e., nonmarket forces) shape the provision and distribution of health care services [21]. Arrows' seminal paper starts with a discussion of why perfectly competitive markets are useful for helping to organize an economic system and then suggests why healthcare is a special case that cannot be left to the market. He notes problems with information will cause market failure, misallocate resources, and so fail to promote economic efficiency in the healthcare sector. We discussed these and other problems toward the end of Chap. 1 and suggested that government (i.e., a nonmarket force) might step in to help things along. The manual process of economic appraisal of healthcare decisions can benefit the economy and society in place of poorly functioning competitive markets.

Health economics is the application of the discipline of economics to the topic of health and health care. An influential health economists, Alan Williams, became involved in this subdiscipline in the early 1970s. He subsequently wrote an expanded definition of health economics in 1987 [22] that included eight topics, which are summarized in Fig. 5. This is known as the plumbing diagram.

The plumbing diagram illustrates that health economics embraces a number of topics, that many of these are interrelated, and that health economists collaborate with other professions (e.g., doctors, politicians, and administrators) and academics (e.g., epidemiologists, statisticians, and bench scientists) [23].

N. Graves et al., *Economics and Preventing Healthcare Acquired Infection*,
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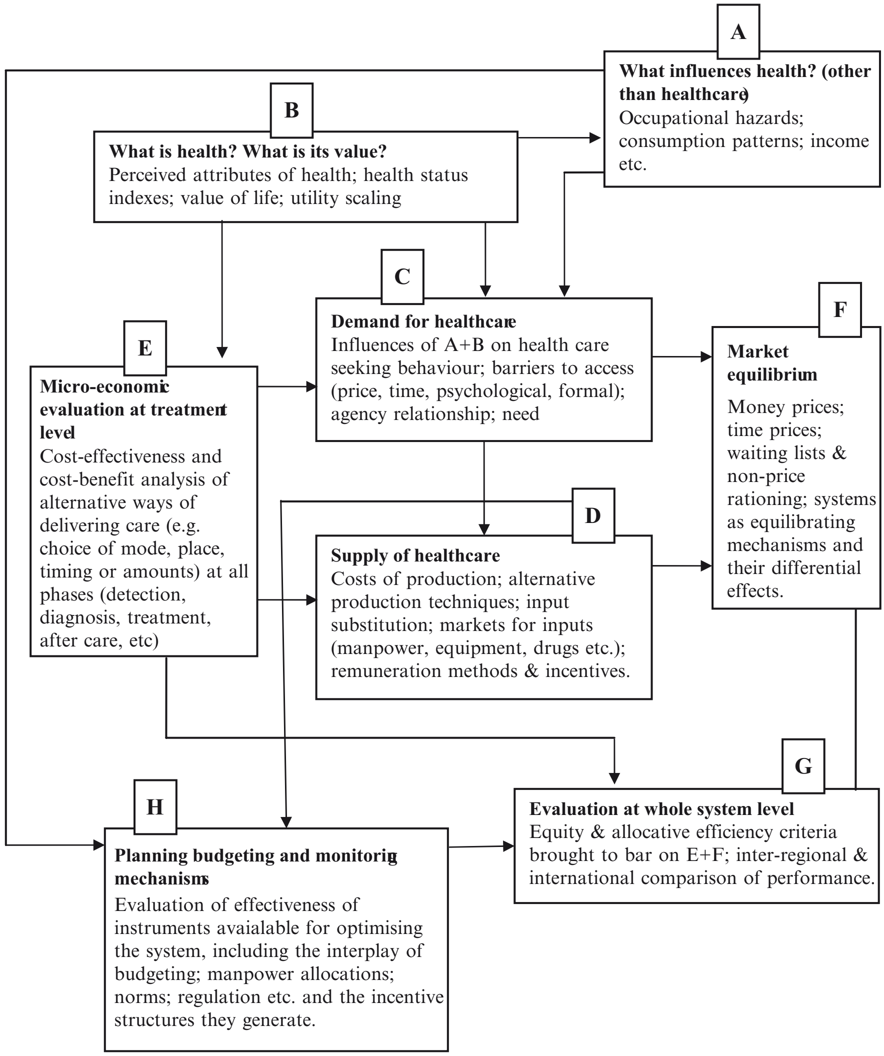


Fig. 5 Alan Williams' description of health economics

Box A is about factors other than healthcare interventions that influence health. Lifestyle choices like smoking and obesity are important determinants of health, as are education, poverty, ethnicity, and the physical environment. There are powerful arguments to be made about changing some of these factors – such as poverty and education – and so reducing the need for invasive and expensive health care services later in life. An example is using economics to inform policy on the regulation of tobacco advertising.

Box B is about measuring and valuing health outcomes. The RAND insurance experiment spawned the now widely used SF-36 health survey [24] that describes both physical and mental health and provides a quantitative score of health outcome across multiple dimensions of health. John Brazier and colleagues are health economists and devised a method for revaluing SF-36 data into a single health-related quality of life score between zero and 1, where zero is dead and 1 represents good health [25, 26]. The outcome measure might seem crude, but it is very useful for weighting the quality of additional years of life arising from some healthcare intervention. The "quality adjusted life year" (QALY) is the result and is used in health economics to describe the benefits of healthcare interventions in terms of how much health gain is produced by a decision to allocate resources to a certain health program. We describe and use the QALY later in this book. Infection control that prevents HAI will cause improved quality of life and may even reduce risk of death and so extend life.

Box C is about the demand for health care. This is related to the determinants of health and how health is measured and valued (i.e., Boxes A and B), since an individual's demand for health care depends on their state of health and how they value improvements in their state of health. The amount of health care any individual demands will also be governed by the price of health care, including nonfinancial costs over and above any professional fees and treatment costs, such as the loss of time it takes to access health care, and psychological factors such as fear of diagnosis and potential poor prognosis. Also, individuals have to use agents (e.g., doctors, health authorities, insurance companies) to demand healthcare, and these agency relationships add further complexity to the demand for healthcare. A nice case study is to imagine whether a patient newly admitted to hospital is able to adequately express their demand for infection control. It is unlikely they will be aware of the latest CDC guidelines or relevant research about risk and infection. For example, few patients would know there is increased risk of catheter colonization if a central line is inserted into the femoral vein rather than the subclavian [27]. The information deficit suffered by the patient illustrates the need for agents to act on their behalf when planning infection control services.

Box D is about the supply of healthcare and the resources used for the production of healthcare. Health economists might investigate why doctors earn 20 or 30 times more than highly skilled nurses in some health systems, explore how capital equipment, such as surgical robots, might substitute or complement human surgeons, and how drug companies might be coerced into lowering prices for life saving therapies. Infection control is one production process used in healthcare. Analyzing the resources used for infection control and how costs change with the scale and scope of infection-control activities falls under this heading.

Box E is about the microevaluation of healthcare programs, and this is another term for "economic appraisal" that was discussed at the end of Chap. 1 and the beginning of this chapter. The activity requires the assessment of the costs and the benefits of alternate ways of using scarce health care resources. It can be used to provide information that would have emerged from perfect markets, if these markets could be used to allocate scarce resource efficiently, which of course they cannot. The endpoint of an economic appraisal is some recommendation about whether a decision to use health care resources in one way over another is likely to improve efficiency within the healthcare sector. Given what we know about scarcity and the need to choose between competing uses of healthcare resources, microevaluation/economic appraisal is potentially powerful. If one healthcare intervention (e.g., MRSA screening) is shown to be better value for money than another (e.g., a surgical robot), and we know we cannot afford to do both, economic appraisal can be used to inform the decision about choosing between them. The microevaluation of healthcare programs is the most visible activity undertaken by those working with health economics because it can inform decision making directly. Many evaluation studies are undertaken each year, and they are often published in good quality medical journals such as JAMA and the New England Journal of Medicine. The methods are often used by noneconomist researchers. Many health care professionals such as doctors and nurses perform and publish economic evaluations. These microevaluations inform both the demand and supply sides of the health care industry. We spend a good amount of this book discussing how economic appraisal can be used for infection control.

Box F is about balancing the demand for health care with the supply of health care. One objective might be to find an efficient allocation of resources (remember economic efficiency from Chap. 1). The role of microevaluation (Box E) on the demand side (Box C) and the supply side (Box D) in finding an efficient outcome for the health care sector is consolidated by Box F.

Box G is about evaluating the entire health care system in terms of efficiency, equity, and cost containment. Crude international comparisons are often made on variables such as proportion of GDP devoted to health care production and life expectancy. For example, we know that US spends more than anyone else on health care but the Japanese live the longest.

Box H is about the assessment of policies that are used to regulate a health care system such as taxes and subsidies and other forms of intervention.

Health economics is a broad topic that can be complex, yet it is often intuitive and easy to interpret. To meet the aims of this book, we need to identify the parts of health economics that are most useful for the infection-control professional.

2.2 The Parts of Health Economics Most Useful for Infection-Control

An infection-control professional – like everybody else – will always face problems of scarcity, that is, wants always exceed means. You may want to implement any number of programs that will directly reduce risk of infection (i.e., use of antimicrobial catheters) or indirectly reduce risks of infection (i.e., staff awareness and education for hand washing), yet you do not have a sufficient budget or enough resources to do everything. You must choose what to do with your scarce resources (i.e., your staff and cash budgets). You should aim to be efficient and get the greatest benefit from your scarce resources. Exactly this question was the subject of a symposium at the Interscience Conference on Antimicrobial Agents and Chemotherapy meeting in San Francisco in 2006, and these were the topics of the symposium:

- Robert Sheretz spoke about "The Most Cost-Effective Way to Spend $100,000 in Infection Control: Investment in Personnel and Education"
- Craig Coopersmith spoke about "The Most Cost-Effective Way to Spend $100,000 in Infection Control: Investment in Infrastructure and Technology"
- Jacques Schrenzel spoke about "The Most Cost-Effective Way to Spend $100,000 in Infection Control: Investment in Microbiology"
- Sanjay Saint asked the question, "Is Infection Control Cost-Effective?"

In the absence of market mechanisms to allocate resources for infection control activities, economic appraisal can be used. You may wish to allocate more resources to infection-control and away from some other activity in the hospital such as out-patient services or cardiology services. You may want to reallocate your existing resources to some other use, such as away from staff education and toward prospective surveillance. You will improve your chances of achieving a reallocation if you make the economics clear for decision makers, and for this purpose, economic appraisal is useful.

There are many examples of how economic appraisal has been used to inform decisions about how to allocate scarce resources. One famous example of a micro-evaluation of a healthcare program (see Box E) was published in 1975 by Neuhauser and Lewicki [28]. They illustrated that testing a stool sample six-times to diagnose colorectal cancer would cost an extra $47 million per case detected. The first five tests were much cheaper per case detected. If a decision were made to allocate resources to six-times testing of stool samples then there would be a few, very grateful patients whose cancer was diagnosed early and cured. The relevant question is what else has been foregone from the millions of dollars allocated to the true positive diagnoses found from the sixth test? Think back to the discussion of opportunity costs in Chap. 1? It is likely that many more lives could have been saved by allocating resources to other competing activities such as expanding vaccination programs, funding drug and alcohol education in schools, and HIV prevention programs.

The economics of a proposal made by a group Duke University cell biologists is another interesting case [29]. They found the incidence of pressure ulcers among hospitalized patients had been reduced but not eliminated despite the use of air-fluidized beds and other specialty devices. They argued that in the future, high risk patients may be sent to space clinics to recuperate in zero gravity for extended periods. Although an appealing solution for a cell biologist, it makes little sense to an economist. We know space travel with existing technology is expensive (see Fig. 2 in Chap. 1), and so the opportunity costs of this decision would be enormous. The resources tied up by providing therapy in the Earth's orbit could be used more productively in other parts of the terrestrial health care system.

Economic appraisal is potentially useful in the healthcare sector as a method of promoting efficiency in resource allocation. To undertake an economic appraisal, we need to use techniques from some of the other boxes in Williams' plumbing diagram. For instance, we will need to know how to estimate the costs of HAIs and the costs of infection control programs. This will use Box D, which is about how healthcare is produced. We will also need to know how to measure and value health outcomes, this will use Box B. We also need to think about the microevaluation at treatment level, the subject of Box E. A number of other disciplines must be involved: clinical expertise is required and we will have to use some epidemiology and statistics. The process of economic appraisal is genuinely multidisciplinary. Although health economics is about more that just economic appraisal (Box E), for now we focus on these techniques. They are powerful and potentially useful for infection control professionals. We describe the different approaches to economic appraisal next.

2.3 Competing Approaches to Economic Appraisal

There are two broad schools of thought about economic appraisal in healthcare and these compete head to head. Competent users of economics should be aware of the differences. The distinctions are often glossed over in papers that discuss the application of economic evaluation for healthcare. First is the w*elfarist* approach that only includes cost-benefit analysis (CBA). Second are *extra-welfarist* approaches that include cost-effectiveness analysis (CEA) and a special form of CEA, cost-utility analysis (CUA). One type of economic appraisal, cost-minimization analysis (CMA), is not useful for decision making in healthcare [30]. A failure to discriminate between welfarism and extra-welfarism could lead to mistakes in the choice of the method used, a failure to apply the method properly and a failure to correctly interpret and report findings. The different methods and the two schools of thought are summarized in Fig. 6.

The advocates for either approach measure the *costs* and *benefits* of competing healthcare interventions to make decisions that promote efficiency. However, despite the apparent similarities, irreconcilable differences exist.

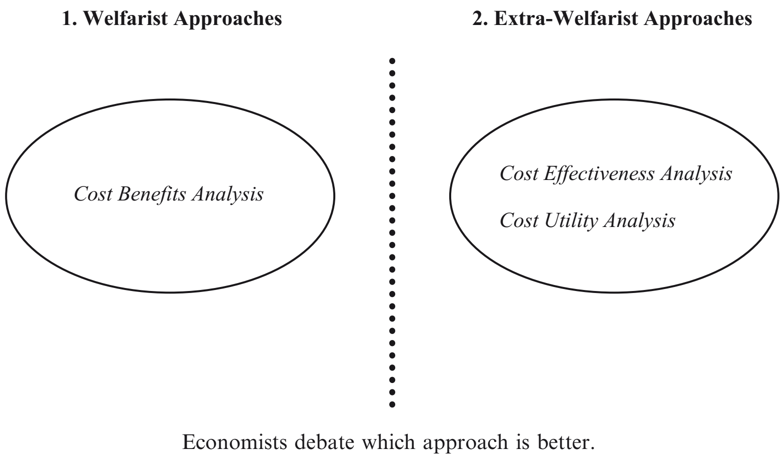


Fig. 6 Welfaist and extra-welfarist approaches to economic appraisal

2.3.1 Welfarism

Welfarists believe that the benefits of a health care intervention should be valued by individual consumers in the same way that any other goods and services are valued. This implies that a consumer will choose the types of healthcare, alongside other goods and services that provide the maximum personal benefit. Individuals are thought to be the best judges of what makes them happy. Welfarists focus on human behavior. If a welfarist sat down on a Sunday evening and planned their weekly shop (i.e., their allocation of resources), it might read something like this:

Things I will buy this week (given scarce resources)

- Dozen Apples, five oranges, and two kiwi fruit
- Cheese balls (packet of 12)
- Dog food
- Calcium supplements to reduce bone loss and risk of fracture in the future
- New socks
- Diet and exercise program to mitigate risk of cardiovascular event
- Beastie Boys album
- Visit path lab for MRSA swab prior to visiting sick relative in ICU
- Cinema tickets
- Repair sole on work shoes

Things I will do without this week (given scarce resources)

- Olive oil (5 L)
- Have test for prostate cancer
- Replace the bald tire on the car
- Visit dentist

This way of thinking arises from the notion of competitive markets, the price mechanism, and the invisible hand, which were all discussed in Chap. 1. Welfarists assume people can find out what healthcare they want and are able to reveal their individual valuation of the benefits of healthcare relative to all other uses of scarce resources. The result is that they choose to consume just the right mix of goods and services to maximize their utility or happiness. We can summarize this argument by a review of the data in Table 3.

The cost of the items are presented in the second column, the dollar valuations of the benefits of consuming these items are presented in the third column and the net benefits, the benefit less the cost (or the profit consumers enjoy), in the fourth. We have sorted these items by the size of the net benefit, so those items consumers value highest are at the top of the list. To be efficient and so gain the most from our scarce resources, we maximize the sum of the net benefits. If our income is \$250 per week, then we work down the list until all our income is allocated between the items. The \$250 is exhausted after we have consumed items 1–10, as follows:

$$\$75 + \$25 + \$10 + \$50 + \$12 + \$10 + \$15 + \$10 + \$5 + \$38 = \$250$$

Net benefit is also maximized:

$$\$25 + \$10 + \$5 + \$5 + \$5 + \$3 + \$4 + \$4 + \$3 + \$3 = \$67.$$

Any reallocation of our resources to include items 11–14 will reduce the total of net benefits and so lead to a less efficient outcome. The reason is that the net benefit of items shaded in gray is less than the minimum net benefit of any of the items not shaded. You can confirm this by substituting any of items 11–14 for any items 1–10 (but remember you have only got $250 to play with). If we took number 2 (calcium + Vitamin D) and number 9 (Cheese balls) out of our list and replaced them with number 11 (Olive Oil), then we have still spent $250 but have reduced out net benefits to $56, and we have achieved less from the same resources. The efficient outcome is to buy items 1–10 and none of items 11–14.

We know that markets fail in health care and so the information summarized in Table 3, which would have arisen so easily in a perfectly competitive market, must be collected manually via economic appraisal. To use economic appraisal, we therefore need a method to identify the costs and benefits of all the available programs. Without this information, it is difficult to find an efficient allocation of scarce resources. Welfarists believe this information can be provided via one form of economic appraisal called cost-benefit analysis. Those who conduct cost-benefit analyses for healthcare decisions try to understand the consumer's preference (i.e., dollar valuation) for health care, just as would have been revealed in a competitive market for sneakers, food, or clothing. This preference is expressed by a dollar valuation of the benefits of healthcare (i.e., the third column in Table 3.). We made a comment in Chap. 1 that valuing the benefits of health care in dollar

Table 3 The cost, benefit and net benefit of possible consumption alternatives

Items that we have valuedw	Cost	Benefit[a]	Net benefit
1. New diet/exercise program to mitigate risk of cardiovascular event	$75	$100	$25
2. Begin a course of calcium + vitamin to reduce bone loss and risk of fracture	$25	$35	$10
3. Beastie Boys new album	$10	$15	$5
4. Visit path lab for MRSA swab prior to visiting sick relative in ICU	$50	$55	$5
5. Dog food	$12	$17	$5
6. New socks	$10	$16	$3
7. Cinema tickets	$15	$19	$4
8. Dozen apples, five oranges, and two kiwi fruit	$10	$14	$4
9. Cheese balls (packet of 12)	$5	$8	$3
10. Repair sole on work shoes	$38	$41	$3
11. Olive oil (5 L)	$30	$32	$2
12. Replace the bald tire on the car	$80	$82	$2
13. Visit dentist	$250	$251	$1
14. Have test for prostate cancer	$500	$501	$1

[a]Benefit is the personal utility gained by consumers and is expressed in terms of their willingness to pay for benefits, relative to all other alternatives

terms is difficult and briefly described the methods at the end of the chapter. Health economists tend to use “contingent valuation” and “choice experiments” to find monetary values for health care programs, but these methods are not widely used and have been criticized [20].

An alternative to welfarism is extra-welfarism, and those who believe this approach is best, advocate the use of two different types of economic appraisal for the purpose of allocating resources, cost-effectiveness, and cost-utility analyses.

2.3.2 Extra-Welfarism

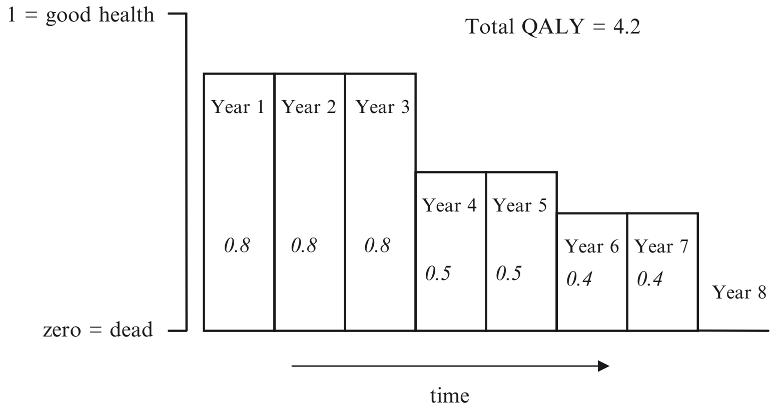
Extra-welfarists debate the assumption that individual consumers of health care should aim to maximize their personal satisfaction, as they would if consuming sneakers, food, or clothing. Extra-welfarists believe resources should be allocated to pursue the social objectives set by the social decision maker. In a health care context, the primary objective of the social decision maker is to maximize the total health of the population. The difference between welfarism and extra-welfarism is that one believes in maximizing personal satisfaction (i.e., happiness or utility) and that this can be expressed in dollar terms, to allow cost-benefit analysis, and the other believes in maximizing health, with all its dimensions such length of life, pain, mobility, anxiety, usual activities, and ability to self-care. The idea of individual consumer sovereignty and the competitive market is downplayed by the extra-welfarists. Extra-welfarism takes a paternalistic approach, tasking government, and other external groups to make recommendations about what healthcare people need, how it should be produced and who gets it. Decision making about how scarce resources are used is taken away from individuals and handed over to government or quasi-government agencies who try to maximize health from scarce resources. The justification is that the decision maker.

> *“occupies his position by virtue of a socially approved political process. He has been entrusted with the task of making choices on behalf of the general public, and this trust implies that he will formulate objectives for the society.” [31]*

The tools of extra-welfarism are cost-effectiveness analysis and cost-utility analysis. The analyst will measure the costs of competing healthcare interventions in monetary units and measure the benefits in terms of changes to health outcomes, not dollar values. An analyst undertaking a cost-effectiveness analysis will characterize health benefits in natural units of health outcome such as life years gained, pain free days, or nosocomial infections prevented. A cost-utility analysis is a special type of cost-effectiveness analysis with health benefits measured by quality adjusted life years (QALYs). See Panel 2 for an introduction to QALYs. This measure of benefit is quite different from the monetary valuation sought by welfarists and allows decision makers to maximize *health* from scarce resources.

For the purposes of this book, we choose to be extra-welfarists and so think about maximizing health benefits as measured by QALYs.

A QALY is derived by measuring the increased duration of life from an intervention and then adjusting the extra years by quality weights. These weights take values between zero (dead) and 1 (good health).



A patient without treatment faces certain death. Luckily for them, they receive a treatment that extends their life by seven years. Over the course of the seven years their quality of their life however will reduce.

during years 1 to 3 the value of the health state they occupy is 0.8
during years 4 and 5 the value is 0.5
during years 6 and 7 the value is 0.4

The total QALYs gained from the treatment is:
0.8 + 0.8 + 0.8 + 0.5 + 0.5 + 0.4 + 0.4 = 4.2 QALYs.

Had they been in good health, they would have enjoyed 7 QALYs.

The treatment generates health benefits valued at 4.2 QALYs.

Panel 2 The quality adjusted life year (QALY)

2.4 Advantages and Disadvantages of Each Type of Economic Appraisal

The welfarist method of cost-benefit analysis is attractive because it is designed to measure what matters to individuals. It accounts for the preferences of individuals for health alongside all the other things they might consume with scarce resources such as cheese balls, dog food, and socks. Because of the compensation principle described in Chap. 1, where the winners from a decision do not actually have to compensate the losers from a decision, a conceptually simple approach to maximizing benefits from scarce resources emerges. Cost-benefit analysis, however, requires individuals to attach a monetary valuation to benefits (see the data in Table 3) and this throws up two problems. One, the wealthy individuals are likely to be willing to pay more for health than the poor, so they may bias resource allocating decisions. Two, it is awkward to ask people to attach a monetary value to health outcomes and gathering the data summarized in Table 3 is a difficult process.

The extra-welfarist methods of CEA and CUA inform the narrower goal of maximizing health outcomes (e.g., measured by QALYs if CUA is chosen) from a fixed pot of resources, such as an annual budget for healthcare expenditures. The power is placed with a notional healthcare decision maker. This method is perceived to be easier to implement than CBA, but does suffer from some criticism, see Coast [32] and Donaldson [33]. A major criticism is that the "decision maker" may not be clearly identified. In reality, decisions are made at many levels and many time points by many different people each with varying objectives that may or may not represents societies objectives. Indeed the Coast paper, published in the BMJ [32] stimulated a lively debate between the welfarists and the extra-welfarists, which can be found on the BMJ Website for those who are interested in a good academic debate.

For the rest of this book, we are going to be extra-welfarists and aim to get the most health benefit from scarce resources to improve efficiency in the health care system. We believe this is closest to the situation faced by those who decide what to do with scarce healthcare resources that may be made available for infection control. It also sidesteps the need for difficult valuations of the benefits of infection control and because infection impacts on both quality and quantity of life, we think the QALY is a suitable outcome measure.

2.5 Conclusions

Health economics is multidisciplinary and broad in scope. The most useful part of health economics for infection control practitioners is economic appraisal. This allows the value for money of competing ways of using scarce health care resources to be assessed. There are two approaches to economic appraisal, welfarist and extra welfarist, and we discussed why we prefer the latter. We choose to describe the costs of health care programs in dollar terms and the benefits in either natural units of output, such as infections prevented or quality adjusted life years (QALYs). Cost-utility analyses – that use QALYs – is the more useful approach to economic appraisal as it allows many different types of health care programs to be compared using a common measure. Cost-utility analysis should be used to evaluate infection control programs.